

Minors on QT

QT-01 Quantum Principles, Structures and Computing

L	T	P	C
3	0	2	4

Prerequisite knowledge: Basic mathematical concepts like Linear equations, differential equations and fundamentals of physics and origin of quantum physics.

Course description and objectives: This course covers the basics of quantum technology and advanced quantum mechanics. It also throws light on fundamentals for quantum computing, and quantum communication.

Module-1

24L+0T+16P = 40 hours

Unit-1

Light – matter interaction; Atomic spectra; Bohr’s quantization principle; coherence, types of coherence -Temporal, spatial, angular; First-Second-order coherence theory; polarization- types of polarization; Interference, interference in Michelson interferometer; Nonlinear optics-Kerr Effect.

Unit-2

Introduction to quantum mechanics, Distinction from classical physics; Heisenberg Uncertainty Principle- implication & applications; postulates of quantum mechanics- quantum states-properties, superposition, entanglement; operators; Eigenvalues; expectation values; Time independent Schrodinger wave equation (Qualitative), Step potential, Rectangular potential barrier, quantum mechanical Tunneling-Applications.

EXPECTED PRACTICES

- ❖ Formation of first-order coherence and interference pattern
- ❖ Measurement of wavelength and coherence length of a laser
- ❖ Determination of coherence time of a light source using a simple interferometer
- ❖ Simulation of Heisenberg uncertainty principle using wave packet evolution
- ❖ Simulation of electron transmission and reflection through a step potential
- ❖ Finding the numerical solution of the time-independent Schrödinger equation for potential well

Module-2

24L+0T+16P = 40 hours

Unit-1

Commutators; Ladder Operators; Number states, Harmonic oscillators (Quantum Vs Classical); Orbital and spin angular momentum; Pauli’s spin matrixes; Hydrogen Atom; Quantum Confinement, Quantum Dots, Quantum Wells.

Unit-2

Phase-space description of a system; Differences between Bosons and Fermions and Properties of photons; Quantum Statistics-Bose-Einstein distributions, Fermi-Dirac distributions; Classical bits vs Quantum bits (Qubits); Superposition; Bell inequalities; Di Vincenzo criteria for realizing qubits; Quantum Computing vs Classical Computing; Q-gates, Universal quantum logic gates; Components of quantum computers, Quantum supremacy and Challenges of quantum computers.

EXPECTED PRACTICES

- ❖ Plot eigenfunctions and energy levels for Quantum Harmonic Oscillator (Simulation)
- ❖ Demonstrate particle-in-a-box energy levels
- ❖ Study quantum dot confinement and energy spacing (Simulation)
- ❖ Show wavefunction overlap and entanglement concept (Simulation)
- ❖ Model classical vs quantum Hamiltonians (Hamiltonian System Simulation)
- ❖ Visualize qubit state vectors and rotations (Bloch Sphere Visualizer)
- ❖ Simulate Bell states and measure correlation (Entangled States and Correlations)

CO No.	Course Outcomes	Blooms level	Module No.	Mapping with POs
1	Apply the fundamental phenomena of light–matter interaction, coherence, interference, and polarization in classical and nonlinear optics.	Apply	3	1(3), 2(3)
2	Apply postulates of quantum mechanics to describe quantum states, operators, and foundational principles like superposition and entanglement.	Apply	3	1(3), 2(3)
3	Analyze quantum systems such as the hydrogen atom and harmonic oscillator; distinguish between bosons, fermions, and photons.	Analyze	4	1(3), 2(3), 4(3)
4	Compare classical and quantum information concepts; apply quantum logic gates and understand the components and challenges of quantum computers.	Analyze	4	1(3), 2(3), 5(2), 11(2)

Text books:

1. Introduction to Quantum Mechanics, David J. Griffiths, Cambridge University Press, 3rd edition, 1018.
2. An introduction to quantum computing P. Kaye, R. Laflamme and M. Mosca, (Oxford Univ Press, 2010)

References books:

1. Introduction to Quantum Mechanics, Griffiths D. J., 3rd Edition, Cambridge University Press (2024)
2. Introduction to Electrodynamics, Griffiths D. J., 4th edition, Cambridge University Press (2020)
3. Principles of Quantum Mechanics, Shankar, R., 2nd edition, Springer (2014)
4. Quantum Mechanics for Engineers and Material scientists, P. Anantram and Daryoush Shiri, World scientific publishing company, 2023.
5. Chris Bernhardt, Quantum Computing for Everyone, The MIT Press, Cambridge, 2020.
6. David McMahon-Quantum Computing Explained-Wiley-Interscience, IEEE Computer Society (2008)
7. Quantum Information Science – Manenti R., Motta M., 1st Edition, Oxford University Press (2023)
8. Quantum computation and quantum information – Nielsen M. A., and Chuang I. L., 10th Anniversary edition, Cambridge University Press (2010)
9. Chuck Easttom, “Quantum Computing Fundamentals”, Addison-Wisley, 2021.
10. Giuliano Benenti, Giulio Casati and Giuliano Strini, “Principles of Quantum Computation and Information”, Volume I: Basic Concepts, “World Scientific”, 2004
11. Leonard Susskind and Art Friedman, *Quantum Mechanics: The Theoretical Minimum*, Basic Books, Perseus Books Group 2015. ISBN-13: 978-0-465-06290-4
12. Eleanor G. Rieffel & Wolfgang H. Polak, *Quantum Computing: A Gentle Introduction*, The MIT Press (Scientific and Engineering Computation series), 2011. ISBN-13 978-0262015066

Online Tools:

IBM Quantum Experience: <https://quantum-computing.ibm.com>

Qiskit Textbook: <https://qiskit.org/textbook>

QuTiP (Python QM simulator)